

Analyzing the Performance of Logical Consistency Tracking among Conversational AI



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PROBLEM DEFINITION & MOTIVATION

Modern conversational AI systems can track user goals (via Dialogue State Tracking, DST) but often fail to adapt when users revise the underlying logic of their queries. Logical consistency tracking — updating reasoning chains when users change assumptions or priorities — remains underexplored.

We aim to explicitly model dynamic logical transitions in conversations, bridging the gap between goal tracking and reasoning evolution. Improving this capability will enhance AI systems' adaptability, coherence, and real-world robustness.

EXAMPLE:

USER: "THE BOY IS PLAYING ON A PEDESTRIAN BRIDGE."

LATER: "ACTUALLY, IT'S PART OF A MAJOR HIGHWAY."

→ A MODEL MUST REVISE THE LOGIC: NOT TREATING THE BRIDGE AS SAFE FOR PEDESTRIANS, BUT RECOGNIZING IT AS A HIGH-TRAFFIC AREA REQUIRING CAUTION.

DATASET CONSTRUCTION

TARCED

- reconstructed factual statements from TARCED
- generated dialogues demonstrating both successful and failed logical transitions

DialogStudio

- curated and reformatted DialogStudio dialogues from MULTIWOZ2.2
- selecting high-quality, multi-turn conversations for manipulation

SNLI

- extracted single sentences as dialogue scenario
- applied prompt-tuned ChatGPT generation to create coherent, logically evolving dialogues

Random Word Pairs

- randomly combined restaurant-related subjects, relations, and objects into triplet
- generated dialogues designed for sequential logical reasoning and revision

Logic Modification and Dataset Assembly

- We built a semi-automated pipeline to create datasets with both **sample** correct and incorrect logic transitions, storing them efficiently in .jsonl files for scalable evaluation. (Correctness isn't fixed)

Datapoint example:

```
"dialogue_id": 0,
"original_context": [
  {"speaker": "User",
   "text": "I am a farmer and want to improve my tomato production. My farm is in a wet and humid climate. What should I focus on first?"},
  {"speaker": "System",
   "text": "In a wet and humid climate, preventing excessive moisture and ensuring proper drainage are key. Tomatoes are prone to fungal diseases ..."},
  {"speaker": "User",
   "text": "I haven't checked. How do I know if my soil drains well enough?"},
  {"speaker": "System",
   "text": "You can test soil drainage by digging a hole, filling it with water, and observing how long it takes to drain. Over 4-6 hours indicates poor ..."},
  {"speaker": "User",
   "text": "Yes! What tomato varieties are best for humid climates?"},
  {"speaker": "System",
   "text": "Fungal-resistant varieties like Defiant PhR, Mountain Magic, or Big Beef are recommended."},
  {"speaker": "User",
   "text": "Since my farm is already humid, how should I water my tomatoes efficiently?"},
  {"speaker": "System",
   "text": "Water only when the topsoil feels dry and use drip irrigation to avoid fungal infections."}
],
"user_revision": [{"speaker": "User",
  "text": "Wait! I made a mistake—my farm is actually in a hot and dry climate, not a humid one. Can you adjust everything for that?"},
  "correct_revised_response": {"speaker": "System",
    "text": "In a hot and dry climate, focus on improving soil moisture retention, not drainage. Choose heat-tolerant varieties like Solar Fire or Heatmaster. ..."},
  "wrong_revised_response": {"speaker": "System",
    "text": "Thanks for the clarification! In hot and dry climates, it's still a good idea to prioritize drainage to avoid any pooling around the roots. Use fungal..."}],
"logical_shift": {
  "original_logic": [{"Climate = humid", "Focus on drainage and air circulation", "Use fungal-resistant varieties", "Water only when topsoil is dry", ...}],
  "correct_revised_logic": [{"Climate = hot and dry", "Focus on water retention and deep roots", "Use heat-tolerant varieties", ...}],
  "wrong_revised_logic": [{"Suggests drainage is a priority in dry climates instead of water retention", ...}]
}
```

EVALUATION FRAMEWORK

- Task: Evaluate GPT-3.5, GPT-4, T5, LLaMA2, Claude on logical consistency after user revisions.
- Stage 1: Model generates a revised response after user updates.
- Stage 2: Extract logic points from model response (via GPT-4).
- Stage 3: Score = (aligned logic points) / (total logic points).
- Stage 4: Automated scoring with structured GPT-4 evaluation prompts.
- Metric: Average score per model across all datapoints.

Metric Trends: Higher logical consistency scores were strongly correlated with correct handling of multi-hop reasoning under user revisions.

RESULTS & ANALYSIS



We explore the challenge of logical consistency tracking in multi-turn dialogue by constructing a benchmark of logic-shifted conversations from DialogStudio, SNLI, TACRED, and random word pairs. We evaluate state-of-the-art models, including GPT-4, Claude, LLaMA2, and T5, using an automatic scoring framework based on extracted logic points. Finally, we investigate the impact of fine-tuning on improving logical adaptation and discuss limitations in current evaluation methods.

Dialogue Length	T5	GPT-3.5	GPT-4	LLaMA2	Claude
Short	0.25	0.65	0.75	0.63	0.84
Moderate	0.25	0.69	0.78	0.68	0.86
Long	0.28	0.69	0.73	0.62	0.82

Short: 1-3 exchanges, Moderate: 4-6 exchanges, Long: 7 or more exchanges.
(1 exchange = 1 user query +1 AI response)

- Claude achieved the highest average score (0.84), followed by GPT-4 (0.76) and GPT-3.5 (0.67), while T5 had the lowest average score (0.25).

Common Errors Observed:

- Failure to update old information.
- Contradictions between old and new logic.
- Over-reliance on surface-level semantic matching without deep reasoning.

FINE-TUNING EXPERIMENTS & FUTURE WORK

Objective: Improve logical adjusting capability for T5, LLaMA2, and ChatGPT.
Results:

- T5 showed very limited improvement in different training parameters.
- LLaMA2 improvement was limited, with persistent inconsistency.

Challenges:

- Default Fine-tuning loss computation focuses on token similarity, not logic adaptation, limiting true reasoning updates; Small dataset (60 dialogues) restricted diversity and generalization.

Next Steps:

- Expand data diversity and explore other loss computation functions & methods to penalize answers with incorrect logic adaptation.

CONCLUSION

To summarize the findings of this study:

- Logical consistency tracking is a distinct capability beyond static goal tracking, requiring models to update reasoning chains in response to user revisions.
- Through dataset construction and evaluation, we observe that even strong models like GPT-4 and Claude show occasional failures in dynamic logic adaptation.
- Fine-tuning on logic-shifted dialogues leads to modest but measurable gains for conversational models, suggesting that targeted supervision can strengthen reasoning coherence across turns.

RELATED LITERATURE

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[2] B. Ghosh, S. Hassan, N. A. Arafat, and A. Khan, "Logical Consistency of Large Language Models in Fact-Checking," 2024.

[3] Y. Zhang, Y. Zhong, D. Chen, G. Angeli, and C. D. Manning, "Position-aware Attention and Supervised Data Improve Slot Filling," Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2017.
[4] J. Zhang, K. Qian, Z. Liu, S. Heinecke, R. Meng, Y. Liu, Z. Yu, H. Wang, S. Savarese, and C. Xiong, "DialogStudio: Towards Richest and Most Diverse Unified Dataset Collection for Conversational AI," EACL 2024 Findings, 2023.

[5] J. D. Williams, A. Raux, and M. Henderson, "The Dialog State Tracking Challenge Series: A Review," Dialogue & Discourse, vol. 7, no. 3, pp. 4-33, 2016.
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[6] J. E. Gonzalez and I. Stoica, "LLMs Can Easily Learn to Reason from Demonstrations: Structure, not content, is what matters!," arXiv:2502.07374, 2025.